Smart City Traffic Intelligence

Part 1 – Data Analytics Insights Report

1. Introduction

This report analyses the Metro Interstate Traffic Volume dataset to understand traffic patterns and the relationship between weather conditions and traffic volume. The dataset contains 48,204 records and 9 original columns, covering traffic volume, temperature, weather conditions, holidays and date/time information.

SQL, descriptive statistics, probability calculations and Power BI were used to analyse the data and identify useful findings for traffic management.

2. Annual Traffic Trends

The SQL analysis compared total recorded traffic volume from 2012 to 2017.

Total recorded traffic increased from approximately 8.2 million in 2012 to 28.2 million in 2013. It then decreased in 2014 and 2015 before increasing again to 29.5 million in 2016 and 35.4 million in 2017.

However, the number of records differs between years. For example, 2015 contained 4,373 records compared with 9,306 in 2016. Therefore, changes in yearly totals may partly reflect differences in data coverage rather than actual traffic growth.

The Power BI dashboard also showed that traffic volume changes throughout the day. Traffic is generally lower during early morning hours and higher during daytime periods.

3. Descriptive Statistics and Correlation

The average traffic volume was 3,259.82 vehicles per recorded hour, with a median of 3,380. The standard deviation was 1,986.84, variance was 3,947,533.43, and range was 7,280 vehicles.

These results indicate considerable variation in traffic volume between different hours.

The correlation coefficient between temperature and traffic volume was 0.1303, indicating a weak positive relationship. This suggests that temperature alone is not a strong indicator of traffic volume. Correlation does not establish causation because other factors, such as commuting schedules, can influence traffic demand.

4. Probability and Congestion Analysis

Congestion was defined as traffic volume exceeding 5,500 vehicles.

The probability of congestion was 14.73%, while the probability of clear weather was 27.78%. The probability of congestion and clear weather occurring together was 3.66%.

Among congested records, 24.83% occurred during clear weather, while 26.30% occurred when temperatures exceeded 292 Kelvin.

The independence analysis compared 0.0366 with 0.0409, suggesting that congestion and clear weather were not independent in the observed data. The odds ratio for congestion

during clear versus cloudy weather was 0.7354, indicating lower congestion odds during clear conditions. These findings show an association, not a causal relationship.

5. Holiday and Weather Analysis

For New Year's Day, average temperature increased from -6.1°C in 2016 to -3.1°C in 2017. Average traffic volume also increased from 1,815 to 2,128 vehicles. No data was available for New Year's Day 2015.

For Labor Day, average temperature decreased from 22.2°C in 2015 to 17.5°C in 2017, while traffic volume fluctuated rather than following a consistent trend.

The Power BI weather analysis showed that Clouds had the highest average traffic volume at 3,618.45 vehicles, while Squall had the lowest at 2,061.75 vehicles. The difference was 1,556.70 vehicles.

6. Data Quality and Dashboard Findings

Power Query was used to check data types, missing values and data quality. Three columns were created: Hour, Temperature_C and Traffic_Category.

Invalid temperature readings of 0 Kelvin were excluded from the Celsius calculation. The cleaned average temperature was 8.11°C.

The Power BI dashboard included daily and hourly traffic charts, weather analysis, a temperature scatter plot, three KPI cards and three interactive slicers.

7. Recommendations and Conclusion

Based on the analysis, the Smart City Mobility Team should monitor high-traffic daytime periods and consider encouraging commuters to travel during lower-demand hours where possible.

Weather conditions can be considered alongside traffic patterns, but they should not be used alone to predict congestion.

The findings are limited by uneven yearly data coverage, repeated timestamps and some invalid sensor readings. Nevertheless, the analysis provides a useful foundation for developing Python and machine learning solutions in Parts 2 and 3.